Speech

Hotwords

Context + Prefix

w_1 w_2 w_3 …

c_1 c_2 c_3 …

Audio Encoder

LLM Token Embedding

LLM Token Embedding

Audio Memory *A*

Word Mean

Trie Token

Decoder State

6 Scalars

Text Memory *C*

Projector

Projector

Candidate Query Q

Q

Q

K_A, V_A

Audio Cross-Attn ×4

Text Cross-Attn ×4

K_C, V_C

Multimodal Fusion

Candidate Self-Attn ×8

Emit Score p_{emit}

Base Logits

Dynamic ΔLogit

+

Next Token

Safety Gate

Trie Order

∧

Top-50

∧

Bilingual CTC

(1) Gating

(2) Evidence Fusion

(3) Decoding

Gate

Frozen backbone

Pretrained core

Alignment

Output